



JOB SHEET 3

Operator, Sequence, Flowchart and Pseudocode

1. Learning Outcome

By the end of this lesson, students must be able to:

- Explain the concept and use of operators.
- Apply sequence structures in Java programming.
- Develop algorithms using pseudocode.
- Develop algorithms using flowcharts.

2. Labs Activities

2.1 Experiment 1: The Use of Operators

Time Allocation : 45 minutes

1. Open a text editor.
2. Create a new file and name it `OperatorStudentName.java`.
3. Write the basic structure of a Java program that contains the `main()` function.
4. Enter the following code inside the public static void `main(String[] args)` method.

```
int x = 10;
System.out.println("x++ = " + x++);
System.out.println("After evaluation, x = " + x);

x = 10;
System.out.println("++x = " + ++x);
System.out.println("After evaluation, x = " + x);

int y = 12;
System.out.println(x > y || y == x && y <= x);

int z = x ^ y;
System.out.println("The result of x ^ y is " + z);

z %= 2;
System.out.println("Final result = " + z);
```

5. Run the program you have created and observe the output.

Questions!

1. Explain, the difference between `x++` and `++x`.
2. What is the result of `int z = x ^ y`? Please calculate it manually.

2.2 Experiment 2

Time Allocation : 45 minutes

Case study

Polinema has a rectangular football field. Create a flowchart and pseudocode to calculate the area of the field.

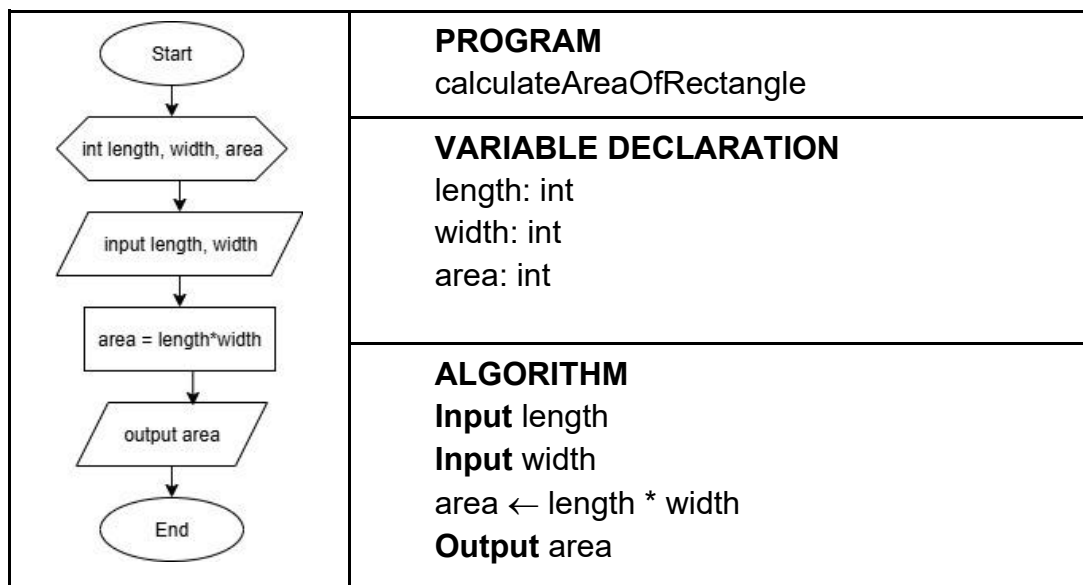
1. Identify **Input**, **Output**, and **Process**

- **Input:** length, width
- **Output:** area
- **Process:**
 - **Read** length, width
 - **Compute** area = length × width
 - **Display** area

2. Identify **Variables** and **Data Types**

Variabel	Data Type
length	int
width	Int
area	Int

3. Implement in Flowchart and Pseudocode



4. Create the program code from the algorithm above, by following these steps:

1. Create a new file and name it `CalculateAreaOfRectangleStudentName.java`
2. Create the basic structure of a Java program consisting of the `main()` function.
3. At first line, add the `Scanner` **library** inside the `CalculateAreaOfRectangleStudentName` class.



```
import java.util.Scanner;
```

4. Declare a `Scanner` object inside the `main()` function.

```
Scanner input = new Scanner(System.in);
```

5. Create the following variables:

```
int length;  
int width;  
int area;
```

6. Get the input value of length and width.

```
System.out.print("Enter the length of the rectangle: ");  
length = input.nextDouble();  
  
System.out.print("Enter the width of the rectangle: ");  
width = input.nextDouble();
```

7. Calculate the area.

```
area = length * width;
```

8. Display the result

```
System.out.println("The area of the rectangle is: " + area);
```

9. Close the `input` object

```
input.close();
```

10. Compile, run the code and observe the result!

Questions!

1. Explain why it is necessary to declare a `Scanner` in experiment 2 above!
2. Explain the purpose of the following program snippet.

```
length = input.nextDouble();  
width = input.nextDouble();
```

2.3 Experiment 3

Time Allocation : 45 minutes

Case Study

Ibu Lani goes shopping at a store and buys clothes priced at Rp. xxx. The store offers a 15% discount on every purchase. Write a program to calculate the discount amount and the total payment Ibu Lani must make.

1. Define input, output, and process

Input: price

Output: discount, totalPayment

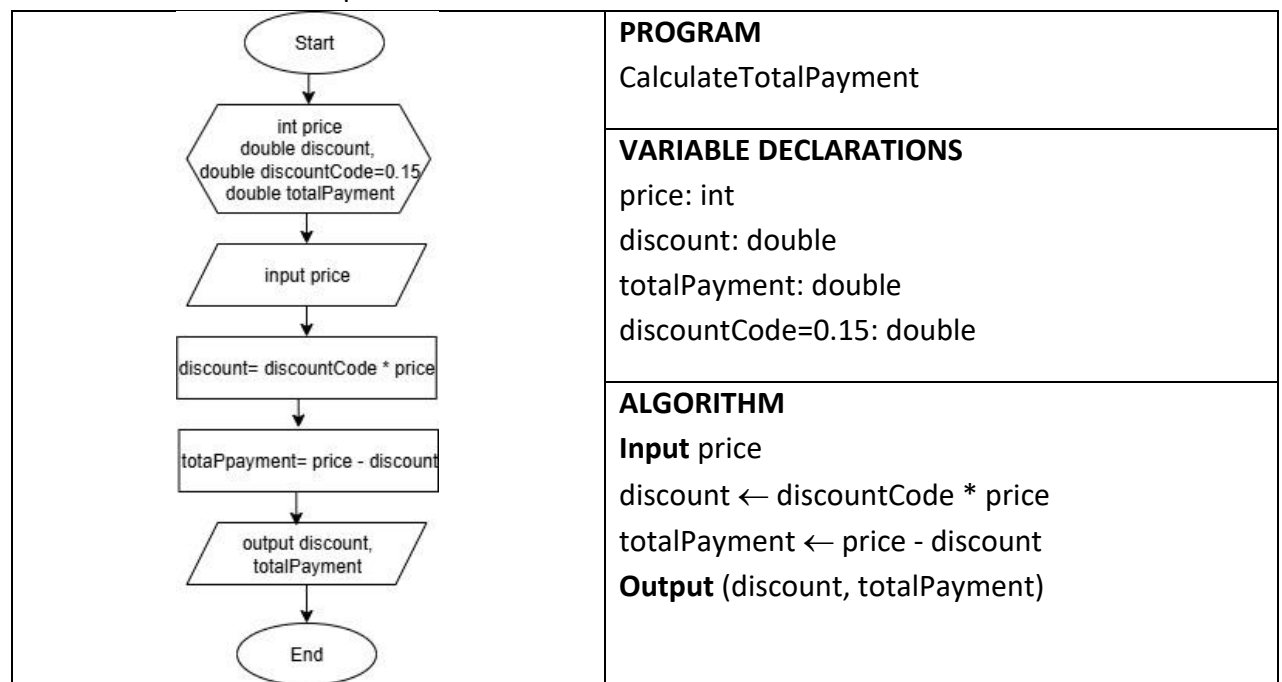
Other data: discountCode=0.15

Process:

1. **input** price
 2. $\text{discount} = \text{discountCode} \times \text{price}$
 3. $\text{totalPayment} = \text{price} - \text{discount}$
 4. **output** discount, totalPayment
2. Identify variable and data type

Variabel	Data type
price	int
discount	double
totalPayment	double
discountCode=0.15	double

3. Create the flowchart and pseudocode





4. Create the program code by following these steps:

1. Create a new file named `CalculateTotalPaymentStudnetName.java`.
2. Build the basic structure of a Java program, including the `main()` function.
3. Add the `Scanner` library within the `CalculateTotalPaymentStudnetName` class.

```
import java.util.Scanner;
```

4. Declare a `Scanner` object inside the `main()` function.

```
Scanner input = new Scanner(System.in);
```

5. Define the following variables:

```
int price;  
double discount;  
double discountCode = 0.15;  
double totalPayment;
```

6. Get the input value of `price`

```
System.out.print("Enter the price of the clothes (Rp): ");  
price = input.nextDouble();
```

7. Calculate `discount`

```
discount = price * discountCode;
```

8. Calculate `totalPayment`

```
totalPayment = price - discount;
```

9. Display the results

```
System.out.println("Original Price: Rp " + price);  
System.out.println("Discount (15%): Rp " + discount);  
System.out.println("Total to be Paid: Rp " + totalPayment);
```

10. Run and observe the result

Questions!

1. Modify the program above by changing the data type of `price` to `double`, then run the program and fix any errors until it executes correctly and produces the expected output.
2. If, in the program above, the total payment is calculated first and then the discount is computed afterward, will the program still run and produce the expected output? Explain your answer.



2.4 Experiment 4

Time Allocation : 45 minutes

Case Study

Employees of PT. Maju Terus receive a basic salary of Rp. x. In addition, each employee is entitled to a transportation allowance of Rp. 600,000, a meal allowance of Rp. 400,000, and a performance bonus equal to 5% of the basic salary. Employees are also required to pay an income tax of 10% of the basic salary. Write a program to calculate the bonus and the total annual salary received by an employee of PT. Maju Terus.

1. Define input, output, and process!

Input: basicSalary

Output: bonus, totalSalary

Other data = transportAllowance = 600.000, mealAllowance=400.00

Process:

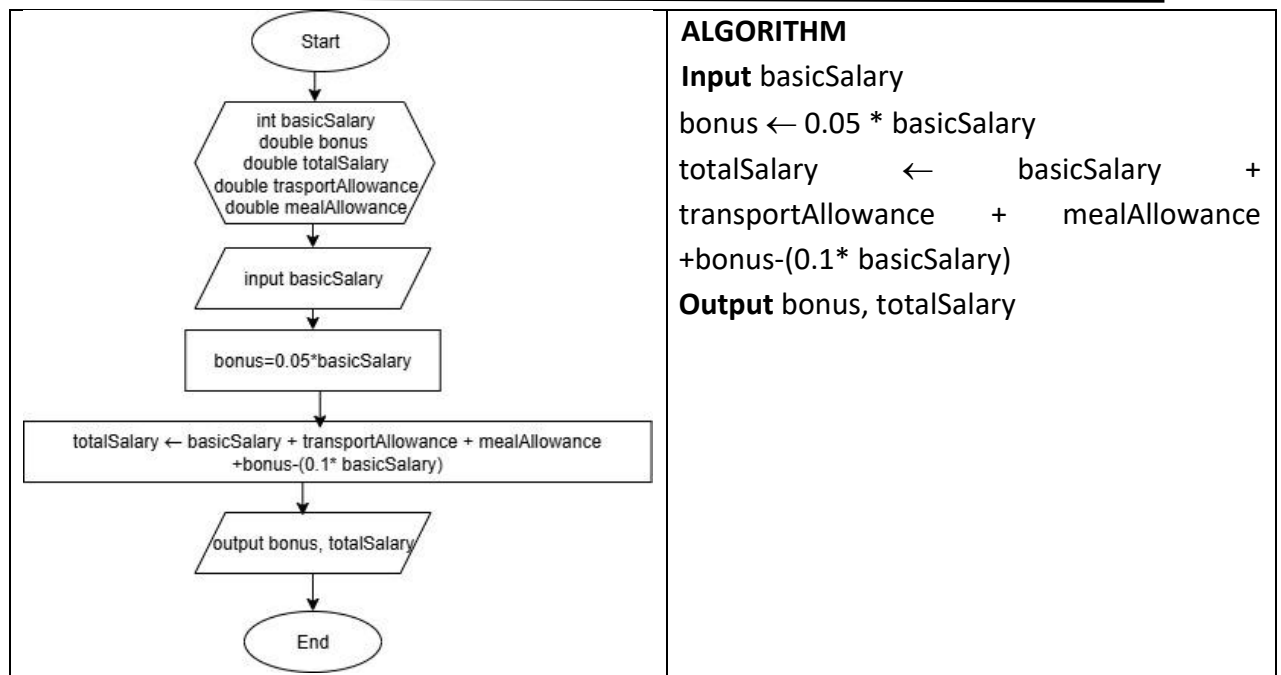
1. Input basicSalary
2. Calculate $\text{bonus} = 0.05 \times \text{basicSalary}$
3. Calculate $\text{totalSalary} = \text{basicSalary} + \text{transportAllowance} + \text{mealAllowance} + \text{bonus} - (0.1 \times \text{basicSalary})$
4. Output bonus, totalSalary

2. Identify variable dan data type!

Variabel	Data Type
basicSalary	Int
bonus	double
totalSalary	double
transportAllowance =600000	Int
mealAllowance =400000	Int

3. Create the Flowchart and Pseudocode!

	PROGRAM Salary
	VARIABLE DECLARATION basicSalary: int bonus, totalSalary: double transportAllowance=600000: double mealAllowance=400000: double



4. Transform the algorithm above into program code!

1. Create a new file named `TotalSalary.java`.
2. Build the basic structure of a Java program consisting of the `main()` function.
3. Add the `Scanner` library in the `TotalSalary` class.

```
import java.util.Scanner;
```

4. Declare a `Scanner` object inside the `main()` function.

```
Scanner input = new Scanner(System.in);
```

5. Create the following variables:

```
int basicSalary;
double transportAllowance = 600000, mealAllowance = 400000;
double bonus, totalSalary;
```

6. Get the input value of `basicSalary`!

```
System.out.print("Enter the basic salary (Rp): ");
basicSalary = input.nextDouble();
```

7. Calculate `bonus`:

```
bonus = 0.05 * basicSalary;
```

8. Calculate `totalSalary`:

```
totalSalary = basicSalary + transportAllowance + mealAllowance + bonus - (0.1 * basicSalary);
```

9. Display the results!

```
System.out.println("\n--- Salary Details ---");
System.out.println("Basic Salary : Rp " + basicSalary);
```



```
System.out.println("Transport Allowance : Rp " + transportAllowance);  
System.out.println("Meal Allowance      : Rp " + mealAllowance);  
System.out.println("Performance Bonus   : Rp " + bonus);  
System.out.println("Total Salary      : Rp " + totalSalary);
```

10. Compile, run and observe the result!

Questions!

1. In the following code snippet, what will happen if the parentheses () are removed, and does it affect the output result? Explain your answer!

```
totalSalary = basicSalary + transportAllowance + mealAllowance +bonus -  
(0.1* basicSalary);
```

2. Modify the program above so that the total salary output is displayed as an integer value!



3. Assignments

Time Allocation: 120 minutes

1. Pak Ali buys a motorcycle on credit for Rp. x . He makes a down payment of Rp. y , and the remaining amount is paid in installments over z months with a fixed interest rate of 1% per month on the outstanding balance. Write a program to calculate the monthly installment amount that Mr. Ali must pay.
2. A car travels from Malang to Surabaya, covering x km. The car consumes an average of 1 liter of fuel every 2 km. If the fuel price is Rp. 10,000 per liter, write a program to calculate the total fuel cost required for the trip.